# BID3000 EKSAMEN

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# Executive summary

During our exam in in the course BID3000 we aimed to better understand the practical application of data warehousing concepts, ETL process implementation, dimensional modelling, business intelligence visualization and our general analytical capabilities. To do this we choose the dataset “Online retail II UCI”. We selected this dataset because it seemed interesting to work with. At the start of the process, we ran some personal tests in an excel sheet to get a better understanding off the dataset. We used Python for the ETL process, PostgreSQL as our database management tool, PowerBI for visualizations and we used Python in a Jupyter Notebook to show analytics.

Our data analysis gives business insight that could have been used to further help the business grow and expand if this had been recent data. This could support strategic decision making to drive growth and expansion, illustrating the potential impact of data-driven decision-making.

Online Retail II UCI does not have a good position, and by going into the seams of their operations we present them with actionable insights that would help them better their position based on hard facts and visuals rather than guesswork.

Through the analysis, we found that *Online Retail II* operates with a highly concentrated market and a strong dependency on the UK, which generates more than £10M of the total £20M revenue. This heavy reliance on one market represents a significant concentration risk, particularly since the business depends on December sales to stabilize yearly performance. The Pareto analysis shows that 80% of the revenue comes from a small group of countries, and an HHI score of 85.7% confirms a highly concentrated market position.

Customer behaviour analysis revealed severe retention issues: 42% of customers purchase only once, resulting in an estimated £3.2M in lost potential revenue. This reflects a clear Pareto principle—0.8% of customers generate 25% of revenue, demonstrating why identifying and nurturing high-value clients is critical.

Overall the findings suggest that *Online Retail II* needs to diversify its market exposure, enhance customer retention initiatives, and address product-level inefficiencies. By leveraging these insights, the company could strengthen its long-term stability and profitability through data-driven strategies.

# Technical Implementation Overview

## Data Warehouse Design and Architecture

We chose to implement a **star schema design** for our data warehouse. The factors that made us choose this was that star schema minimize the number of joins required for different analysis, improving execution time on queries. When using interactive dashboards this is kind of critical for ad-hoc analysis. And as for the business users, the denormalized structure is intuitive to understand. And as we are using BI tools, star schema enables faster visualizations and responsive dashboards.

As for our **data model structure** it consists of 2 fact tables and 4 dimension tables. The "factsales" table captures all transactions with measures as quantity, unit price and revenue. While the "factcancellations" captures cancelled orders with quantity cancelled and revenue lost.

The **dimension tables** consists of "dimdate", "dimcustomer", "dimproduct" and "dimcountry". *Dimdate* would be the time dimension (year, quarter, month, day, day of week). *Dimcustomer* holds information for customers with support for anonymous customers (CustomerID = 0). *Dimproduct* catalogue holds information about stock codes, descriptions, prices and a shipping indicator flag. *Dimcountry* would be the geographical dimension for internation market analysis.

Both fact tables have the **transaction line-item level** granularity which is the most atomic level available. This ensures maximum analytical flexibility that allows aggregation to any higher level (customer, product, time period) while at the same time makes it possible to go down to individual transactions details when needed.

Some of the principles we used to guide our design was **surrogate keys**, **business process separation** and **data quality by design**. All dimensions use auto-generated int surrogate keys instead of natural keys for improved performance. The natural keys would be retained as attributes for traceability. As for the sales and cancellations in different tables is because they represent distinct business processes. The separation improves query performance and maintains a clear business logic boundaries. As for any invalid data, we filter this during the ETL loading rather than at the query time. Making sure analysis work with validated data. You can see the ERD diagram as an appendix /Documentation/ERD_diagram.pdf

## ETL process implementation

**Extract phase:** We begin the extract phase with the .CSV file containing over 1 million transactions from Dec 2009 to Dec 2011. The files include invoice numbers, stock codes, product descriptions quantities, prices, customer ID and country information.

**Transform phase:** The transformation logic consists of data quality filtering, data standardization and business logic implementation.

The filtering removes some invalid stock codes, transactions with zero quantities, invalid pricing and missing descriptions. These are meaningless and contains bad data.

During the standardization it cleans up some product descriptions using trim operations and title case formatting. NULL customerID are mapped to 0 (anonymous customer entity).

The business logic implementation separates negative quantities into cancellations fact table. It also generates customer and product surrogate keys for dim tables. It also flags some products with “is_shipping=TRUE” based on keywords.

**Load phase** : This phase follows a specific sequence.

-It populates dim tables first to establish FK relationships.

-Then there is a in-memory lookup for optimizing fact table loading.

-Fact tables are inserted with the FK references, uses the in memory to find the right ID fast.

-Then it uses batch commits (saving many instead of one at the time) for improved performance.

The ETL process is implemented in Python using pandas for data manipulation and psycopg2 for PostgreSQL database interaction. Detailed process flow with screenshots is documented in /Documentation/ETL_process_with_key_screenshots.docx and /Documentation/Data_quality_issues.docx

## Analytical query development

SQL analytics implementation

We developed six queries to demonstrate dim modelling capabilities and extract business insights:

1. **Time-Based Analysis:** Year-over-year quarterly revenue comparison by country using LAG window functions with PARTITION BY, revealing Q4 revenue peaks and seasonal patterns
2. **Aggregation Operations:** Revenue aggregation using GROUP BY ROLLUP to show hierarchical totals by country, year, and quarter.
3. **Window Functions:** Product ranking by monthly revenue with 3-month rolling averages using RANK() and AVG() window functions to see trending products.
4. **Complex Filtering:** High-value international customers with cancellation history identified using CTEs, subqueries, and EXISTS/IN clauses.

5. **Business Metrics - Customer Analysis:** Customer lifetime value calculation with RFM-style segmentation (Recency, Frequency, Monetary) using multiple CTEs and CASE-based classification.
6. **Business Metrics - Product Analysis:** Product performance evaluation with custom health score algorithm weighing revenue, customer reach, geographic distribution, and cancellation rate.

Each query includes inline business interpretations as comments. The whole SQL code and findings are available in /Database/queries_bid3000.sql and /Database/Business_interpretation_of_findings.docx

Key SQL findings were:

- UK alone stands for 92% of the revenue.
- Q4(quarter 4) generates 38% of annual revenue.
- Top 50 customers makes 25% of the total revenue and that
- The star products" with health score above 85 points generate 42% of revenue.

## Python analytics and machine learning

**Data warehouse integration**

Python analytics connect to the PostgreSQL data warehouse to extract transaction level data for modelling. This analysis implements descriptive as well as predictive analytics approaches.

**Classification model: High Value Customer Identification**

We developed a random forest classifier to identify high value customers based on transaction behaviour. We were very happy to see the performance on this one with very high accuracy and precision.

**Accuracy:** 96% classification accuracy

**Precision:** 96% for high-value customers

**Recall:** 97%

**F1-Score:** 96%

**AUC:** Near-perfect area under ROC curve

Out of 1,149 customers tested, the model made only 22 mistakes—less than 2% error rate. This means the business can trust the model to identify valuable customers for special treatment and marketing campaigns.

The analysis showed that purchase quantity is by far the most important factor (90% importance). This means that customers who buy more items are more valuable than customers who buy expensive items.

**Business implications**: The classification model proves reliable for applications such as customer segmentation and targeted retention campaigns. This means that you can use the model to identify the best customers for "special treatment". And use the CLV model to create a priority liste on whom to call first, second third etc. Even though it cannot predict exactly revenue.

Complete Python code and detailed statistical analysis are available in /Analytics/BID3000.ipynb with additional business interpretations in /Analytics/Findings.docx.

## Business Intelligence Dashboard

**Dashboard architecture**

For this exam we developed an interactive PowerBI dashboard to provide visual analytics and business insights. The dashboard consists of three main pages that each serves distinct analytical purposes. The main dashboard page provides KPI-cards (total revenue, total orders, average order value, revenue lost to cancellations, cancellation rate percent), revenue trends by time, geographic revenue using a logarithmic scale, top 5 selling products and cancellations trend.

The detailed analysis page enables interactive filters by year, quarter, month and country. It also includes cross-filtering capabilities and an Azure map visual showing revenue distribution globally.

The customer segments page (advanced analytics) translates machine learning insights into doable recommendations. It would tell the sales team who to contact and which actions to take with them. The customers are grouped into 7 different categories from "champions" (best customers) to "lost" (inactive customers). It helps the sales team on what to offer the different kinds of customers.

Advanced features: The dashboard provides us with vast amounts of advanced DAX measures. Users can click on any element on the chart to filter everything else, for example by clicking "Germany" shows only German data across the charts. The use of blue as the primary colour is because of its associations with trust, calmness, stability, reliability and the fact that people are found to be more productive in blue rooms (Savavibool, 2018).

The complete Power BI file is available at /Dashboard/PowerBI.pbix with screenshots in /Dashboard/PowerBI_screenshots/ and a user guide at /Documentation/PowerBI_dashboard_Brief_Userguide.md

## Key Insights and findings

The main business operations are happening in UK, and this is currently the primary market focus by far. This one single country is also the primary revenue driver at >£10 M of a total revenue of ≈£20M. This is a massive concentration risk, especially considering the reliance of December to save the fiscal year each year. The Pareto diagram shows 80% of revenue comes from a small set of countries, and the Herfindahl-Hirschman Index shows 85.7% which is indicative of a highly concentrated market.

We found that most transactions are of relatively low value (~£20) as revenue is highly right-skewed, and that there is an overall downward trend in revenue, which is propped up only by the December rush and the rare occasion of bulk orders from whales in Thailand and Singapore. 2010Q4 are never surpassed in 2011-2012. Peak revenues decline YoY. The deterioration is confirmed through the lack of higher highs and higher lows in the line chart in the PowerBI dashboard. ≈30% of the product revenue comes from a cake stand.

### Revenue

- Revenue is highly right-skewed. This means that most transactions are of low value. KPI shows average order value is £20.
- UK accounts for most of the revenue, £10M out of £20M.
- Netherlands shows the highest median transaction in the box-and-whiskers plot. This shows potential for higher ROI growth and where to concentrate on next.
- Denmark and Spain show high upper-bound values which is indicative of deals of the type of the Singaporean and Thai whales.
- Core transaction range manifests at 10-100 units and £10-£100 revenue
- Sparse high-quantity/high-revenue points
- 2010Q4 peaks never surpassed. No sustained growth.
- Lack of higher highs and higher lows confirms deterioration
- A 3-tier cake stand dominates revenue share. Literally a third of it.
- Gift/party/home-décor customers
- UK>£10M is a massive concentration risk.
- The combined revenue of the next best markets, Netherlands, Germany and France is £1.2-£1.5M

### Customers

There are practically three distinct revenue driving customer groups. There is the one who consists of many customers who buy small once or seldom, then there's the few customers who buy bulk. 30% of customers have a 15-month inactivity (Silhouette analysis S0). Then there's the Potential Loyalists and the Loyalists who have respectively 27.3% of base, and 34.8% of base with 38.6% of the revenue. In the Order Frequency vs. Total Spend we found four customer segments. 1: High-frequency + high-spend, these are the core revenue drivers. Next, 2: High-spend, low-frequency, the occasional bulk buyer customer. Then there's 3, the mid-frequency and mid-spender, which have growth potential. Lastly at 4, there's low-frequency low-spend, which have a high churn risk and low ROI to reactive. Still, the long tail of low-engagement customer in the scatter plot shows low-engagement customers dominates the base.

- ≈80% of revenue comes from a small set of countries where the UK is dominant
- $R^2$=71.2% gives 83.8% of predictions within ±20% of CLV which is acceptable for business decisions.
- Model performs best for those who are not high-CLV customers, but omitting the high-CLV ones just artificially heightens the R2 and gives a misrepresentation.
- The elbow plot confirms 4 clusters as optimal
- The F value (frequency) drives M value (monetary), with R (recency) showing weak correlation.
- For the silhouette analysis, it shows K=5 as near optimal because better interpretability than K=7.
- Segments:
    - S0 – At Risk (≈30% of customers, 15-month inactivity)
    - S1- Champions (7% of base, 29.7% of revenue)
    - S2 – Champions (whales) (14 customers, 16.4% revenue)
    - S3 – Potential Loyalists (27.3% of base, growth opportunity).
    - S4 – Loyalists (34.8% of base, 38.6% of revenue)

## Business Recommendations

Investigate the “worst performing products”. Some products have a very high cancellation rate, and business owners need to know why, or how to improve, or maybe take the products out of the inventory. Improve “manual order” process. Implement order verification checklist for manual entries. Add mandatory quality check before order confirmation. Train staff on accurate product information. This to reduce cancellations down from 37.84% down to more manageable percentage. Add more info about products with a high cancellation rate. Maybe with more details about products, dimensions, materials and user instructions.

Get more valuable feedback from customers. An automated survey could be sent out to customers that have cancelled something. Have them answer questions about why they cancelled, reason behind it when they contact the customers from using the Action Rank column in the Customer Segments page in the PowerBI file. Then analyze the feedback for systematic issues.

Use machine learning to identify “high risk” orders. Flag those orders and make sure to have a pro-active customer service contact with those customers, could potentially save a lot of revenue over the year. Reduce general cancellation rate down from 1.84% --> 1.0% could make out £450 000 and improve customer satisfaction and brand reputation.

**Optimize International Market Performance**

As we see on the data, UK dominates the revenue (92% total), meaning that the rest of the countries have lots of untapped potential for upping the revenue.

Start with Germany (largest EU economy) but also focus on the other “big ones” as France, Netherland and Spain with a lot of potential. Have some good marketing campaigns, maybe collaborate with some local influencers for brand awareness.

**Maximize Customer Lifetime Value**

If the business could convert 10% of the one-time buyers into repeating customer, it would generate additional revenue up to £2M and that would be a good lift in the economy.

**Upselling**

Based on the top selling products analysis, the business could make a bundle, selling all the top matching products with a discount and implement a recommendation engine that shows on product pages and checkouts. So it would be easy for potential customer to add new things into the basket as they proceed to checkout.

**Prioritize seasonal holidays**

We can see in our analysis that Q4 generates a whopping 38% of the total revenue, with a strong peak in Nov-Dec. It could be wise to increase inventory for top 20 products to avoid stockouts and capture a full Q4 demand for optimal revenue.

Make sure to preorder seasonal items from suppliers. Make sure to have a regular stock monitoring to avoid stockouts and have a rapid reordering chain. If the business could reduce stockouts by 90% it should improve revenue by additional £800k.

After the holidays there should be a clearance sale to reduce inventory, and free up some cash and warehouse space. If the business could reduce excess inventory by 30% it should give £200K cost savings. Also a good customer service 24/7 could be a good way to improve branding and reduce cancellations.

**Other things to consider:**

If there are products with <10 sales a year it should be removed from catalogue and free up warehouse space. Maybe discontinue the bottom 10% of the products and focus more on the top 20% products for increased revenue.

## Team member contribution

We have been working in the form of daily physical meetings with Lucas working mainly remotely with meetings over Teams as he lives in Oslo, but he also joined us physically, and in general have had tight communication over the "messenger" app. This form makes who have done what a bit fuzzy and we have also overlapped, so in terms of contributions the following are the gist of it. There are no freeloaders, this would make Kristian jump ship stat like in the first semester, and it is why he has stayed with this group the other semesters as well. So, in the big picture,

Jonas has mainly worked on the initial data cleaning of the dataset and the general ETL process, He has also done large parts of the python analytics integration and has written large our Jupyter Notebook file. He also contributed towards the SQL queries.

Kenneth has been working on putting together the report, fine tuning the different elements in the report. And trying to have a general overview. Organizing and trying to put down on the board what needs to be done and by when. Been working on business recommendations as well as detective work all over the report trying to find missing pieces, holes etc. Worked on business interpretations of findings as well as data issues identified and how they were handled. Put together the ERD diagram with Kristian. Been writing some of the technical implement overview and business recommendations.

Lucas started out by trying to define the initial setup for the SQL database by defining the different entity relationships and trying to assume datatypes and the use of surrogate keys. Also "helped" by misunderstanding the python analytics part of task two and spent some time writing code for predictive models we were not supposed to use. Then wrote the findings document in the analytics folder, containing business interpretations for the predictive models that we *actually* were supposed to use.

Kristian has mainly been focusing on the Power BI dashboard, the pages who support the dashboard, and the DAX for the measures for the visuals. He has also written the PowerBI_dashboard_Brief_Userguide.md, and taken screenshots of the PowerBI.pbix file, and provided attempts at organizing the file structure and database connection username and passwords in a useful manner because we're still sometimes running the file from "C:/Users/{name}" which obviously gives traceback errors when using relative paths, so now me make sure to have cd'ed to the folder the file actually is in before we run it... 3rd year students or what, huh…? Also wrote the key insights and findings and restrained to write what they should do, because that's for the next title.

The operations for the data have been a major "what should we do?"-nut to solve and every member has contributed significant amount of evaluating how in- and excluding the different columns for different business questions would impact the outcome of the quality of the data to make those business decisions off of.

## Use of AI

We have used AI during different parts of our exam. At the start of the exam, we mainly used it as a sparring partner for parts we wanted to comprehend better to get better control of what we were supposed to create. During our ETL process we used AI (Claude LLM) to help us understand what our dataset consisted of. This helped us better understand the dataset, what needed to be cleaned etc in the data process. We also used AI to help us formulate SQL queries and write some boilerplate python code that we used in our Jupyter notebook & also to double check our code was correct in relation to what statistical models wanted to use. At the end of the exam, we also used AI to help fix grammar and spelling mistakes when writing the final report.

# Reference list

Savavibool, N.; Gatersleben, B.; Moorapun, C. (2018). The Effects of Colour in Work Environment: A systematic review. *Asian Journal of Behavioral Studies, vol 3, no. 13.*